

14. ALPHA-BETA PRUNING

AIM

To implement the Alpha-Beta Pruning algorithm in Python for a Tic-Tac-Toe game, where the computer selects the best possible move efficiently.

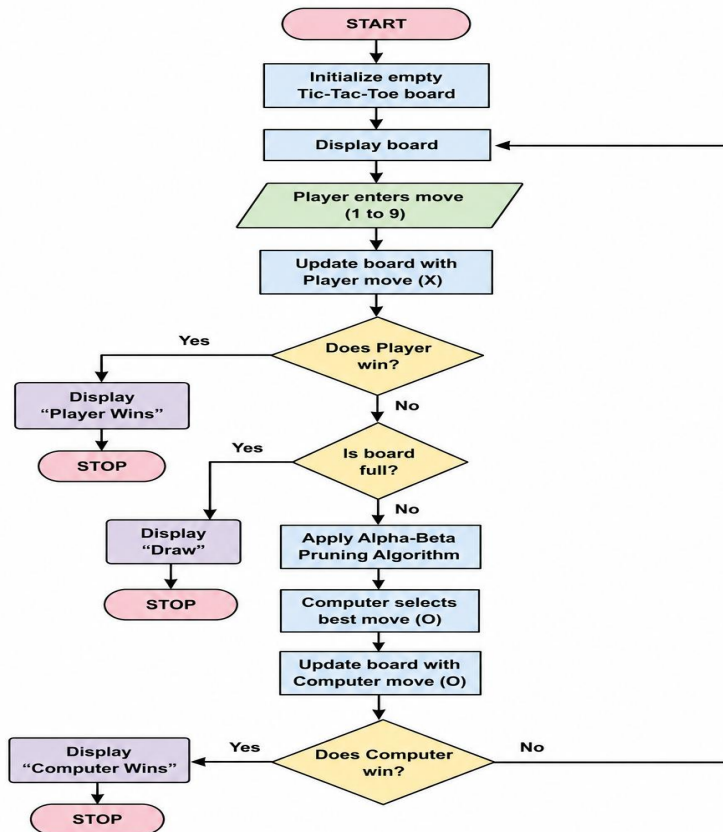
OBJECTIVE

- To understand the Alpha-Beta Pruning algorithm.
- To implement game-playing using Python.
- To reduce unnecessary searches in the Minimax algorithm.
- To determine the winner or draw.

ALGORITHM

1. Start the program.
2. Initialize an empty Tic-Tac-Toe board.
3. Define all possible winning combinations.
4. Accept the player's move manually.
5. Check whether the player has won.
6. Apply the Alpha-Beta Pruning algorithm to find the computer's best move.
7. Update the board with the computer's move.
8. Check whether the computer has won.
9. Repeat the process until a player wins or the board is full.
10. Display the winner or draw.
11. Stop the program.

FLOWCHART:



PYTHON CODE:

```
b=[" "]*9
```

```
w=[(0,1,2),(3,4,5),(6,7,8),(0,3,6),(1,4,7),(2,5,8),(0,4,8),(2,4,6)]
```

```
def win(p): return any(all(b[i]==p for i in x) for x in w)
```

```
def ab(a,be,maxi):
```

```
    if win("O"): return 1
```

```
    if win("X"): return -1
```

```
    if " " not in b: return 0
```

```
    if maxi:
```

```
        v=-2
```

```
    for i in range(9):
```

```
        if b[i]==" ":
```

```

        b[i]="O"; v=max(v,ab(a,be,0)); b[i]=" "
        a=max(a,v)
        if a>=be: break

    return v

v=2
for i in range(9):
    if b[i]==" ":
        b[i]="X"; v=min(v,ab(a,be,1)); b[i]=" "
        be=min(be,v)
        if a>=be: break

    return v

for t in range(9):
    print(b[:3],b[3:6],b[6:])
    if t%2==0:
        p=int(input("Enter position (1-9): "))-1
        if b[p]!=" ": continue
        b[p]="X"
    else:
        best=max((ab(-2,2,0),i) for i in range(9) if b[i]==" ")[1]
        b[best]="O"
    if win("X"): print("You Win!"); break
    if win("O"): print("Computer Wins!"); break
else: print("Draw!")

```

OUTPUT:

```
Output

[' ', ' ', ' ', ' '] [' ', ' ', ' ', ' '] [' ', ' ', ' ', ' ']
Enter position (1-9): 1
['X', ' ', ' ', ' '] [' ', ' ', ' ', ' '] [' ', ' ', ' ', ' ']
['X', ' ', ' ', ' '] [' ', ' ', ' ', ' '] [' ', ' ', ' ', 'O']
Enter position (1-9): 9
['X', ' ', ' ', ' '] [' ', ' ', ' ', ' '] [' ', ' ', ' ', 'O']
['X', ' ', ' ', ' '] [' ', ' ', ' ', ' '] [' ', 'O', ' ', 'O']
Enter position (1-9): 3
['X', ' ', 'X', ' '] [' ', ' ', ' ', ' '] [' ', 'O', ' ', 'O']
Computer Wins!

=== Code Execution Successful ===
```

RESULT:

Thus, the Alpha-Beta Pruning algorithm was successfully implemented in Python for Tic-Tac-Toe. The computer efficiently finds the best move by eliminating unnecessary game states and correctly determines the winner or draw.